

GURU NANAK DEV ENGINEERING COLLEGE, LUDHIANA

Department of CSE

ASSIGNMENT PLAN

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Branch	Semester	Subject	Code	Session
CSE	3 rd	Data Structures	CCS203	July-Dec.2025

Assignment No.-2

Content Covered:

Unit-5 Queues: Sequential and Linked representation of queue, Types of queue- Linear Queue, Circular Queue, Dequeue, Priority Queue, Operations on each types of Queues and their algorithms, Applications of Queues.

Unit-6 Trees: Basic terminology, Sequential and linked representations of trees, Tree Traversals, Different types of Trees- Binary Tree, Binary search tree, Threaded binary tree, AVL tree, B-tree, B+ - tree, Red Black tree. Operations on each of the trees. Application of Binary Trees.

Unit-7 Heaps: Representing a heap in memory, Operations on heaps, Application of heap in implementing priority queue and Heap sort algorithm.

Unit-8 Graphs: Basic terminology, Representation of graphs – Adjacency matrix, Adjacency list. Operations on graph, Traversal of a graph – Breadth first search, Depth first search. Shortest path algorithms - Dijkstra's and Floyd-Warshall algorithm. Minimum spanning tree – Prim and Kruskal. Applications of graphs.

Unit-9 Hashing and Hash Tables: Introduction to hash table, Hash functions, Concept of collision and its resolution using open addressing and separate chaining, Double hashing, Rehashing.

Q1 (A) Predict the output of enqueue and dequeue operations if: Queue = [, ,], front = -1, rear = 1, operations: Enqueue(5), Enqueue(10), Dequeue(), Enqueue(15), Enqueue(20).

(B) Construct a BST for the elements: 50, 30, 70, 20, 40, 60, 80. Show the resulting tree and its in order, preorder, and post order traversals. Compare and contrast Binary Search Tree, AVL Tree in terms of ordering and usage. Construct a B-Tree of level 3 for the set of elements.

(C) Given the array [10, 20, 15, 30, 40], show the tree representation of:

(a) Min Heap

(b) Max Heap.

What will happen if a heap property is violated at multiple levels simultaneously? Explain your answer.

Q2. (A) Consider the following undirected graph:

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A — B — C
|   |   |
D — E — F
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Edges: A-B, A-D, B-C, B-E, D-E, E-F, C-F

(a) Perform BFS traversal starting from vertex A and show the queue contents at each step.

(b) Using BFS, find the shortest path from vertex A to vertex F.

(c) Explain how BFS ensures the shortest path in an un-weighted graph.

(B) A hash table has 10 slots (0–9). Keys are inserted using hash function $h(k) = k \bmod 10$ and linear probing.

Insert the following keys: 10, 20, 30, 41, 22, 31

(a) Show the final hash table.

(b) How many comparisons are required for inserting each key?